Plant Disease Detector

Course: ITAI 1378 Computer Vision
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Project Tier: Tier 1- Image Classification

The Problem

Many plants develop diseases caused by fungi, bacteria or environmental stress. Farmers and home gardens often struggle to identify these diseases early.

Misidentifying a plant diseases can lead to crop loss, wasted treatments and reduced food production.

This problem matters to :

- Farmers
- Home gardeners
- Agricultural business

Early detection can help prevent plant damage and improve crop health.

The Solution

I built a computer vision model that detects plant diseases from leaf images.

The system will:

- Take an image of a plant leaf
- Use a trained CNN model to classify the disease
- Return the predicted disease label

A deep learning image classifier that identifies plant diseases from leaf photos.

Technical Approach

Computer Vision Technique:
Image Classification

Model:

- Convolutional Neural Network (CNN) using ResNet18
- Framework:
- Pytouch

Why this approach:

CNN models are well suited for image classification task because they can automatically detect patterns such as leaf spots, discoloration and texture changes associated with plant diseases

Data Plan

Dataset Source:

PlantVillage Dataset (public dataset)

Dataset link:

https://www.kaggle.com/datasets/emmarex/plantdisease

Dataset Size:

- 50,000 images

Labels:

Examples include:

- Healthy
- Leaf Spot
- Rust
- Powdery Mildew

Data Preparation

- Resize images
- Normalize pixel values
- Split dataset into train/ validation / test

System Diagram

Leaf Image

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Image Preprocessing

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CNN Model

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Disease Prediction

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Output Label

Example Output:

Prediction: Tomato Leaf Spot

Confidence: 92%

Success Metrics

Metric Type	Measurement	Target
Primary	Classification Accuracy	greater than 85%
Secondary	Inference Speed	less than 1 second per image
Additional	Precision/ Recall	greater than 80%

Week-by-Week Plan

Week Task Milestone

Week 10	Download dataset, set up environment	Dataset ready
Week 11	Train CNN model	Model runs successfully
Week 12	Tune parameters, improve accuracy	>80% accuracy
Week 13	Build demo notebook	Demo working
Week 14	Documentation and testing	Repo complete
Week 15	Final presentation	Project demo

Challenges & Backup Plans

Challenge	Plan B
Dataset too large	**Use smaller subset**
Model accuracy too low	Use transfer learning
Training too slow	Train using Google Colab

Resources Needed

Resource	Tool
Compute	**Google Colab**
Framework	PyTorch
Dataset	PlantVillage (Kaggle)
Cost	Free

Final Project Improvements

- Completed implementation of the CNN/ResNet18 model
- Added prediction functionality for plant leaf images
- Generated sample prediction outputs with confidence scores
- Organized GitHub repository with documentation and results
- Created a demo-ready computer vision system

Conclusion

- Successfully developed a plant disease detection system using computer vision
- Demonstrated how CNN models can classify plant diseases from leaf images
- Showed the potential of AI to support agriculture and early disease detection
- Future improvements could include larger datasets and real-time deployment